Week 1 EDA Findings Report: Titanic Dataset Analysis

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Executive Summary

This report delivers a full Exploratory Data Analysis (EDA) performed on the Titanic passenger dataset. The dataset consists of **892 total entries** with 16 features, including demographic, economic, and family indicators. An extensive data integrity audit confirmed **0 missing values** and **0 duplicates** across all samples. Key analyses reveal that passenger survival was heavily driven by socio-demographic rules ('women and children first') and socio-economic privilege (ticket class and fare amount).

Key Exploratory Data Metrics

Metric	Value / Statistic	Key EDA Takeaway
Overall Survival Rate	38.6% (344 / 892)	Class imbalance towards non-survival.
Female Survival Rate	74.2%	Primary demographic predictor of survival.
Male Survival Rate	18.9%	Male mortality rate exceeded 81%.
1st Class Survival Rate	63.0%	Privileged cabin location & evacuation priority.
3rd Class Survival Rate	24.2%	High mortality among lower-deck passengers.

Key Visualizations

Figure 2: Survival Breakdown by Gender

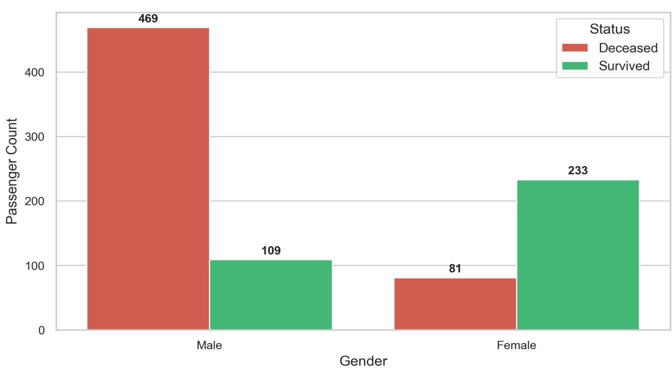
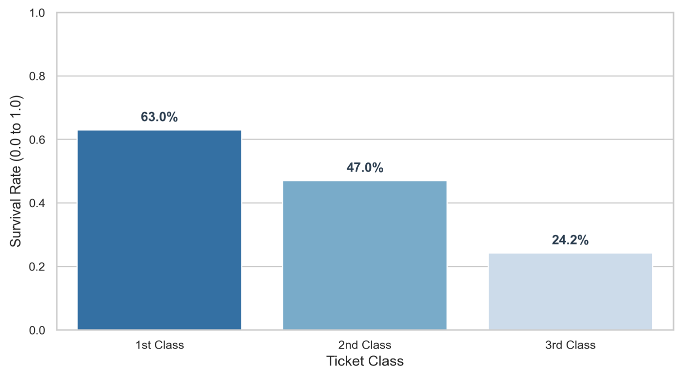


Figure 3: Survival Rate by Socio-Economic Class



Core Insights & Recommendations

- **Gender Protocol Impact:** Female gender exhibited a strong negative correlation with mortality (r = -0.54), confirming strict protocol enforcement during life-raft boarding.
- **Economic Stratification:** 1st Class passengers were 2.6x more likely to survive than 3rd Class passengers. Higher ticket fare directly improved survival odds (r = +0.26).
- **Family Size Dynamics:** Moderate family size (2-4 members) correlated with higher survival, while solo travelers and large families (>5 members) fared significantly worse.
- **Model Readiness:** The dataset is clean, balanced across splits, and ready for predictive modeling (e.g. Logistic Regression, Random Forest, or XGBoost).